THESIS LEARNING SUMMARY

Filtered Vector Search on Proximity Graphs | September 2026

Current status: I have completed and reviewed CMU 15-445/645 Lectures #01, #03, and #04 using lecture summaries, flashcards, quizzes, and mind maps. I am now preparing to move from database fundamentals to vector-search-specific topics.

1. Topics Learned in Depth

Lecture / Resource	Main topics covered	Connection to thesis
CMU 15-445/645 #01 — Relational Model & Algebra	Relational model; relations; tuples; attributes; schema; relational algebra; basic DB organization.	Foundation for understanding how database systems represent and process structured data.
CMU 15-445/645 #03 — Database Storage I	Database storage hierarchy; volatile vs. non-volatile storage; database files; pages; page headers; slotted pages; slot arrays; tuples; Record IDs; variable-length data; overflow storage.	Provides the storage-level foundation for understanding how database and vector-index data are physically organized and accessed.
CMU 15-445/645 #04 — Memory Management	Buffer pool; frames; page table; buffer hit/miss; pin/unpin; dirty pages; page replacement; LRU; Clock; LRU-K; memory–disk interaction; caching.	Directly relevant to memory access cost, caching, CPU-side performance, and later search/traversal optimization.

2. Preliminary Research Topics Investigated (Not Yet Studied in Depth)

Vector Database • Vector Embeddings • ANN/ANNS • Proximity-Graph Search • HNSW • Filtered Vector Search / FANNS • Metadata Filtering • Filtered-DiskANN • ACORN • NaviX • JAG • RACORN-1 • BOA

3. Future Learning Resources — Selected Only

Playlist / Resource	Selected videos / topics	Purpose
CMU Intro to Database Systems (15-445/645 — Fall 2025)	#05 Database Storage II; #08–09 Indexes & Filters I–II; #13–14 Query Execution I–II; #15–16 Query Planning & Optimization	Strengthen DB/index/query fundamentals needed for the thesis.
ML⇄DB Seminar Series — Fall 2023	Qdrant: Vector Search Engine Internals; selected Vector DB system talks (Weaviate / Milvus / pgvector)	Bridge from general DB systems to vector-database systems.
CMU Advanced Database Systems (15-721 — Spring 2024)	#04–05 Query Execution & Processing; #06 Vectorized Query Execution; #07 Code Generation & Compilation; #08 Scheduling & Coordination; #16 Cost Models	Advanced CPU, execution, optimization, and systems knowledge.

4. Next Research Learning Path

Database fundamentals → Indexes & Query Execution → Vector Databases → ANN/ANNS → HNSW → Filtered Vector Search (FANNS) → Existing filtered-search methods → CPU profiling & adaptive traversal.